N5 Pneumatics Worksheet MI

1. Write down the formula that we use to calculate the force in a single-acting cylinder as it outstrokes.

F = P x A

2. A pneumatic stamping machine is used to stamp the company logo onto metal casings. It is discovered that the stamp does not imprint the logo properly. Suggest ways of increasing the size of the force produced by the cylinder.

Increase the surface area and/ or increase the pressure

3. What controls the instroke of a single-acting cylinder?

The tension of the return spring and force acting on the outstroke surface area.

4. A single-acting cylinder is used to press two sheets of acrylic together when they are gluing. The process requires a force of 300 N. The only piston available has a diameter of 20 mm and it is supplied with air at a pressure of 0.3 N/mm^2. Will this arrangement enable this process to be carried out properly?

d = 20mm

P = 0.3N/mm^2

A = πr^2

A = $\pi 10^2$ = 314mm^2

F = PA

F = 0.3 × 314 = 94.2N Therefore answer is **No**

5. A double-acting cylinder has a piston diameter of 25 mm and a rod diameter of 10 mm. Pressure = 0.4 N/mm². Calculate the instroke force.

d_{piston} = 25mm

d_{rod} = 10mm

P = 0.4N/mm^2

A = $\pi r_{piston}^2 - \pi r_{rod}^2$

A = $(\pi \times 12.5^2) - (\pi \times 5^2)$

A = 490.8738521 – 78.53981634

A=412.334mm^2

F = PA

F = 0.4 × 412.334 = **164.9N**

6. What force will be produced by a 20 mm diameter cylinder as it goes positive using a pressure of 0.8N/mm^2?

d = 20mm

P = 0.8N/mm2

A = πr^2

A = $\pi 10^2$ = 314mm^2

F = PA

F = 0.8 × 314 = **251.2N**

7. Calculate the outstroke force produced by a 40 mm diameter cylinder when it operates with a supply pressure of 3bars.

d = 40mm

P = 3 bar = 0.3N/mm^2

A = πr^2

A = $\pi 20^2$ = 1256 mm^2

$F = PA$

$F = 0.3 \times 1256 =$ **376.8N**

8. Write down the formula that we would use to calculate the pressure of a system if we already knew the force required and the size of the cylinder available.

$P = F / A$

9. A double-acting cylinder has a piston diameter of 60 mm and a rod diameter of 10 mm. The air supply pressure is 12bar. Calculate the instroke force.

$d_{piston} = 60mm$

$d_{rod} = 10mm$

$P = 12 \text{ bar} = 1.2N/mm^2$

$A = \pi r_{piston}^2 - \pi r_{rod}^2$

$A = (\pi \times 30^2) - (\pi \times 5^2)$

$A = 2827.433388 - 78.53981634$

$A = 2748.893572mm^2$

$F = PA$

$F = 1.2 \times 2748.893572 =$ **3298.67N**

10. A stamping machine exerts a force of 454N with a piston diameter of 34mm. Calculate the air pressure required for this operation.

$F = 454N$

$d = 34mm$

$P = F/A$

$A = (\pi d^2)/4$

$A = (\pi \times 34^2)/4 = 907.46\ mm^2$

$P = F / A$

$P = 454 / 907.46 =$ **$0.5N/mm^2 = 5bar$**

11. A machine that places tops on bottles uses a single-acting cylinder. The process requires a force of 650N. What air pressure needs to be supplied to the cylinder with a diameter of 56 mm?

$F = 650N$

$d = 56mm$

$A = A = (\pi d^2)/4$

$A = (\pi \times 56^2)/4$

$A = 2461.76mm^2$

$P = F / A$

$P = 650 / 2461.76 =$ **$0.26\ N/mm^2 = 2.6bar$**

12. A force of 540 N is needed to push a packing case off a conveyor belt. The single-acting cylinder used has a diameter of 60 mm. What air pressure should be supplied to the system?

$F = 540N$

$d = 60mm$

$A = A = (\pi d^2)/4$

$A = (\pi \times 60^2)/4$

$A = 2826mm^2$

$P = F/A$

$P = 540 / 2826 =$ **$0.19N/mm^2 = 1.9bar$**

13. A pneumatic system is used to test the quality of drawer guides in kitchen cabinets. A force of 16 N is needed to open the drawer. The single-acting cylinder available has a piston diameter of 10 mm. What air pressure should be supplied?

$F = 16N$

$d= 10mm$

$A = (\pi d^2)/4$

$A = (\pi \times 10^2)/4$

$A = 78.5\ mm^2$

$P = F/A$

$P = 16 / 78.5 =$ **$0.20\ N/mm^2 = 2\ bar$**

14. Write down the formula we would use to find the area of a piston if we already knew the size of the force it needed to produce and the air pressure being supplied.

$A = F / P$

15. A single-acting cylinder is used to lift parcels on to a conveyor. This requires a force of 180 N with the system operating at a pressure of 6bars. Calculate the area of the piston required.

$F = 180N$

$P= 6\ bar = 0.6\ N/mm^2$

$A = F / P$

$A = 180 / 0.6 =$ **$300\ mm^2$**

16. A door requires a force of 400N to slide it open. A single-acting cylinder supplied with a pressure of 5bars controls the operation. Calculate the diameter of the piston required to produce this force.

$F = 400N$

$P = 5\ bar = 0.5N/mm^2$

$A = F / P$

$A = 400 / 0.5$

$A = 800mm^2$

$A = (\pi d^2)/4$

$800 = (\pi d^2)/4$

$4 \times 800 = \pi d^2$

$3200/ \pi = d^2$

$1019.10828 = d^2$

$d = \sqrt{1019.10828}$

$d = 31.923475$

$d =$ **$32.0mm$**

17. A furnace door weighs 100 N and is lifted by a single-acting cylinder as it outstrokes. Compressed air is supplied at a pressure of 4 bars. Calculate the diameter of the piston required to raise the door.

$F = 100N$

$P = 4\ bar = 0.4N/mm^2$

$A = F / P$

$A = 100 / 0.4$

$A = 250mm^2$

$A = (\pi d^2)/4$

$250 = (\pi d^2)/4$

$4 \times 250 = \pi d^2$

$1000/ \pi = d^2$

$318.3098862 = d^2$

$d = \sqrt{318.3098862}$

$d = 17.84124$

$d = \mathbf{18mm}$